FEDERAL SPECIFICATION

STAKES, SHEET METALWORKING; AND PLATE, BENCH STAKEHOLDER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers stakes used by sheet metal workers for shaping and forming metals, and a bench plate for holding the stakes.

1.1.1 *Federal specification coverage.* — This specification does not cover all the types of stakes and plates, but only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.*—Stakes and plate covered by this specification shall be of the following types, as specified (see 6.1).

Stakes:
- Type I.—Beakhorn.
- Type II.—Blowhorn.
- Type III.—Bevel-edge square, with offset head.
- Type IV.—Creasing, with horn.
- Type V.—Hatchet.
- Type VI.—Square.
- Type VII.—Round head.
- Type VIII.—Bottom.
- Type IX.—Double-seaming.
- Type X.—Double-seaming, with four interchangeable heads.
- Type XI.—Hollow mandrel.
- Type XII.—Candle mold.

Plate:
- Type XIII.—Bench.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standard.—The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ–M–151—Metals, General Specification for Inspection of.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design.—Stakes shall be properly proportioned to be strong and durable and shall be of a design suitable for the purpose intended.

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3.1.1 *Illustrations.*—The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of stakes and plates otherwise conforming to this specification.

3.2 Material.—The stakes and plate shall be of the material specified herein for each type.

3.3 Finish.

3.3.1 *Working surfaces.*—All working surfaces, after machining to true shape, shall be polished, using an abrasive of not less than 100 grit in the final operation.

3.3.2 *Other surfaces.*—All nonworking surfaces shall be smooth. Welded fillets shall be ground smooth. All unpolished surfaces shall be coated with a rust-resistant paint.

3.4 Marking.—Stakes and plate shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.5 Stakes.

3.5.1 *Construction.*—Stakes shall be cast or forged as specified herein for each type. Forged stakes shall be forged integral or fabricated by welding.

3.5.2 *Shanks.*—The shanks of all stakes shall be square in section at the lower end and tapered to fit the cored holes in the bench plate specified in 3.6. The shanks of all stakes shall be at right angles to the body or head.

3.5.3 *Type I, beakhorn.*—Stakes shall be of forged steel and shall consist of a body and shank.

3.5.3.1 *Body.*—The body shall be provided with two horns, the tops of which shall be level and in the same plane. One horn shall be circular in cross section at the point of juncture with the shank. The bottom side of the circular horn shall taper gradually in a curve from the shank toward the end. The outer end of the circular horn shall be tapered abruptly from the bottom to the top surface at an angle of approximately 25 degrees from the vertical. The other horn shall be rectangular in cross section with the top surface level and shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale. The bottom and sides of the rectangular horn shall be tapered toward the end.

3.5.3.2 *Shank.*—The shank shall extend from the juncture of the two horns.

3.5.3.3 The stake shall conform to table I and shall be similar to figure 1.

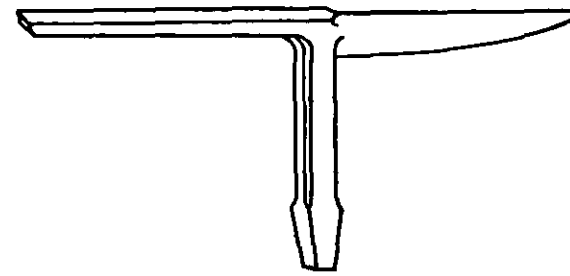


FIGURE 1.—Type I, beakhorn.

3.5.4 *Type II, blowhorn.*—Stakes shall be of forged steel and shall consist of a body and a shank.

3.5.4.1 *Body.*—The body shall be provided with two horns. One horn of the body shall be semicircular in cross section, and the other horn

TABLE I.—*Type I, beakhorn*

Round horn			Flat horn			Weight
Length ±1	Diameter at shank ±¼	Diameter at end ±5/16	Length ±1	Width at shank ±¼	Width at end ±5/16	±3½
Inches 16½	*Inches* 2⅜	*Inch* 15/16	*Inches* 21½	*Inches* 2⅜	*Inches* 1 11/16	*Pounds* 38½

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TABLE II.—*Type II, blowhorn*

Half round horn			Round horn			Weight
Length ±1/8	Diameter at shank ±1/4	Diameter at end ±1/16	Length ±1	Diameter at shank ±1/8	Diameter at end ±1/16	±1 3/4
Inches 9	*Inches* 4 3/4	*Inch* 7/16	*Inches* 18	*Inches* 1 1/2	*Inch* 5/8	*Pounds* 14 1/4

circular in cross section. Both horns shall be tapered from the shank to the ends.

3.5.4.2 *Shank.*—The shank shall extend from the juncture of the two horns.

3.5.4.3 The stake shall conform to table II and shall be similar to figure 2.

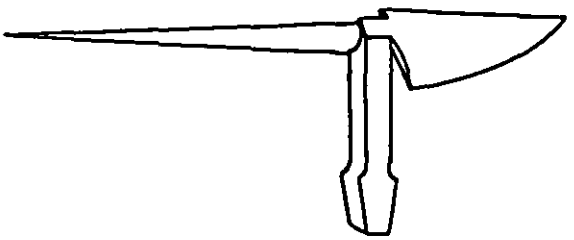


FIGURE 2.—Type II, blowhorn.

3.5.5 *Type III, bevel-edge square, with offset head.*—Stakes shall be of forged steel and shall consist of a flat head and an offset shank.

3.5.5.1 *Head.*—The head of the stake shall be rectangular in shape. The two long sides of the head shall be slightly beveled or tapered toward the shank and the two short sides shall be square with the top face. The working surfaces of the head shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale.

3.5.5.2 *Shank.*—The shank shall be offset in two places at an angle of 90 degrees. The offsets shall be in opposite directions approximately 8 1/2 inches apart.

3.5.5.3 The stake shall conform to table III and shall be similar to figure 3.

3.5.6 *Type IV, creasing, with horn.*—Stakes shall be of forged steel and shall consist of a body and shank.

TABLE III.—*Type III, bevel-edge square, with offset head*

Face		Weight
Width ±5/16	Length ±1/4	±1 1/2
Inches 2 11/16	*Inches* 4 3/4	*Pounds* 14

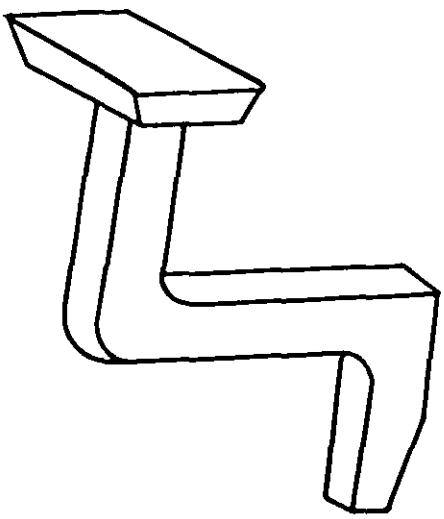


FIGURE 3.—Type III, bevel edge square, with offset head.

3.5.6.1 *Body.*—The body shall be provided with two horns, the tops of which shall be level and in the same plane. One horn shall be rectangular in cross section and shall taper in thickness from the shank to the end. The horn end shall taper from the top to the bottom at an angle of approximately 45 degrees with the vertical. The top of the rectangular horn shall contain six semicircular grooves across the face, ranging in size from approximately 1/8 inch to

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$1\frac{1}{32}$ inch in diameter. The smallest groove shall be at the outer end with the grooves increasing in size toward the shank. The working surface shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale. The other horn of the stake shall be circular in cross section and shall taper in diameter from the shank to the end.

3.5.6.2 *Shank.*—The shank shall extend from the juncture of the two horns.

3.5.6.3 The stakes shall conform to table IV and shall be similar to figure 4.

TABLE IV.—*Type IV, creasing, with horn*

Creasing horn	Round horn			Weight
Length ±⅛	Length ±¼	Diameter at shank ±¼	Diameter at end ±1/16	±1½
Inches 6¾	*Inches* 11¾	*Inches* 1½	*Inch* 7/16	*Pounds* 13½

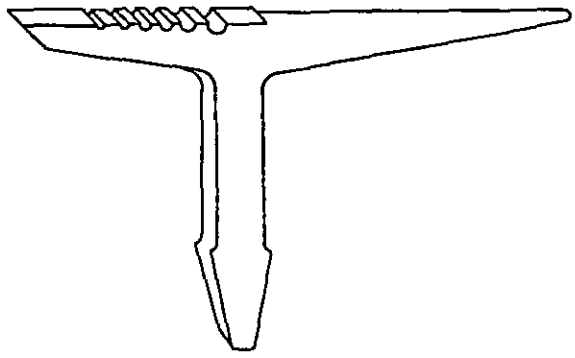


FIGURE 4.—Type IV, creasing, with horn.

3.5.7 *Type V, hatchet.*—Stakes shall be of forged steel and shall consist of a body and a shank.

3.5.7.1 *Body.*—The top of the body shall be beveled to an angle of approximately 45 degrees. The working surface of the body shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale.

3.5.7.2 *Shank.*—The shank shall be located approximately equidistant from both ends of the body with one side of the square flush with the side of the body that forms an angle of approximately 45 degrees with the top of the body.

3.5.7.3 The stake shall conform to table V and shall be similar to figure 5.

TABLE V.—*Type V, hatchet*

Length of blade ±½	Weight ±1
Inches 13	*Pounds* 8½

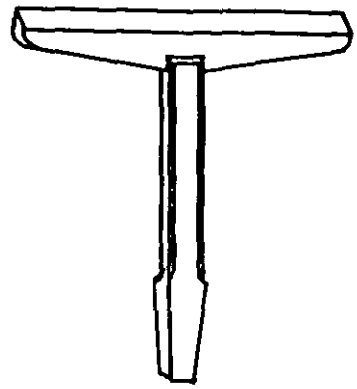


FIGURE 5.—Type V, hatchet.

3.5.8 *Type VI, square.*—Stakes shall be of forged steel and shall consist of a head and a shank.

3.5.8.1 *Head.*—The head shall have a top face rectangular in shape, and shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale. One side of the head shall be beveled toward the shank at an angle of approximately 5 degrees.

3.5.8.2 *Shank.*—The shank shall extend from the center of the head.

3.5.8.3 The stake shall conform to table VI and shall be similar to figure 6. The sizes shall be as specified (see 6.1).

TABLE VI.—*Type VI, square*

Width of face ±¼	Length of face ±¼	Weight ±1
Inches 2½	*Inches* 4½	*Pounds* 11½
3¼	5¼	14

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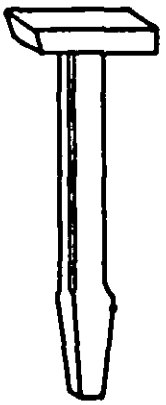


FIGURE 6.—Type VI, square.

3.5.9 *Type VII, round head.*—Stakes shall be of cast iron and shall consist of a rounded head working surface integral with the shank.

3.5.9.1 The stake shall conform to table VII and shall be similar to figure 7.

TABLE VII.—*Type VII, round head*

Diameter of head ±1/16	*Weight* ±1/8
Inches	*Pounds*
3 1/16	10

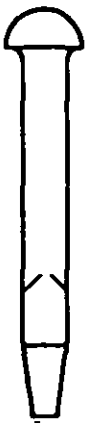


FIGURE 7.—Type VII, round head.

3.5.10 *Type VIII, bottom.*—Stakes shall be of forged steel and shall consist of a head and a shank.

3.5.10.1 *Head.*—The head shall be rectangular in cross section having a rounded top working face with one side straight and the other side tapered and rounded.

3.5.10.2 *Shank.*—The upper part of the shank shall be rectangular in cross section and the lower end square. The working surface of the head shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale.

3.5.10.3 The stake shall conform to table VIII and shall be similar to figure 8.

TABLE VIII.—*Type VIII, bottom*

Width across top ±3/16	Thickness at top +3/64 −1/32	Weight ±1/10
Inches	*Inch*	*Pounds*
1 3/16	5/64	2



FIGURE 8.—Type VIII, bottom.

3.5.11 *Type IX, double-seaming.*—Stakes shall be of forged steel and shall consist of a body and a shank.

3.5.11.1 *Body.*—The body shall be composed of two horns. The tops of the horns shall lie in the same plane. The end of one horn shall be beveled toward the shank. The end of the other horn shall be rounded. The ends of the horns shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale.

3.5.11.2 *Shank.*—The shank shall extend from the juncture of the two horns.

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3.5.11.3 The stake shall conform to table IX and shall be similar to figure 9.

TABLE IX.—*Type IX, double-seaming*

Diameter of each horn ±5/16	Length of longer horn ±¾	Length of shorter horn ±⅝	Weight ±4
Inches 1 3/16	*Inches* 17	*Inches* 12⅛	*Pounds* 40

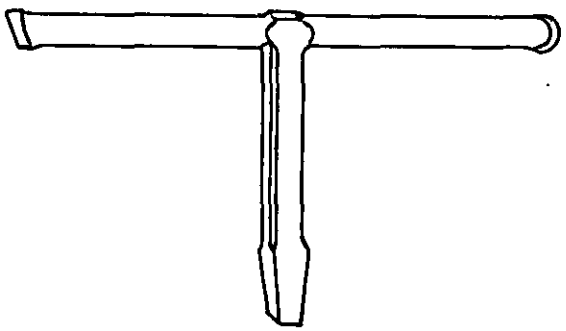


FIGURE 9.—Type IX, double-seaming.

3.5.12 *Type X, double seaming, with four interchangeable heads.*—Stake shall be of cast iron and shall consist of a body and four detachable heads.

3.5.12.1 *Body.*—The body shall be square in cross section and provided with two tapered shanks at right angles to each other at one end so that the stake may be used in either a vertical or horizontal position. The body shall have two sockets of the same size at right angles to each other at the other end to receive any one of the four heads.

3.5.12.2 *Heads.*—The four interchangeable heads shall be one each of round, long oval, square, and short oval.

3.5.12.3 The stake body and heads shall conform to table X, and shall be similar to figure 10.

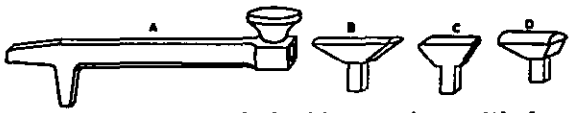


FIGURE 10.—Type X, double-seaming, with four interchangeable heads.

3.5.13 *Type XI, hollow mandrel.*—Stakes shall be of cast iron and shall consist of a body and a clamping bolt.

3.5.13.1 *Body.*—The body shall be provided with two horns. One horn shall have a flat rectangular working surface, with beveled sides. The other horn shall be semicircular in cross section. The body shall have a full length "T-slot" on the underside.

3.5.13.2 *Clamping bolt.*—A "T" head clamping bolt with lever nut shall be provided for fastening the stake to the bench.

TABLE X.—*Type X, double-seaming, with four interchangeable heads*

		Round head (A)		Long oval head (B)			
Length over-all ±½	Weight ±12½	Height[1] ±⅛	Diameter ±¼	Height[1] ±⅛	Length ±⅝	Width at wide end ±¼	Width at narrow end ±¼
Inches 30½	*Pounds* 107½	*Inches* 4¼	*Inches* 6	*Inches* 2½	*Inches* 8⅜	*Inches* 3	*Inches* 2½

Square head (C)			Short oval head (D)			
Height[1] ±⅛	Length ±¼	Width ±⅝	Height[1] ±¼	Length ±¼	Width at wide end ±¼	Width at narrow end ±¼
Inches 2½	*Inches* 5½	*Inches* 4½	*Inches* 2⅝	*Inches* 5¾	*Inches* 4	*Inches* 3¾

[1] Measured from top of head to shank.

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TABLE XI.—*Type XI, hollow mandrel*

Over-all length ±2	Weight ±7	Flat horn		Rounded horn	
		Length ±1¼	Width ±⅝	Length ±2	Radius ±⅛
Inches	*Pounds*	*Inches*	*Inches*	*Inches*	*Inches*
40	49	0	5⅝	31	2
60	110	11¼	6⅛	48	2½

3.5.13.3 The stake shall conform to table XI and shall be similar to figure 11. The size shall be as specified (see 6.1).

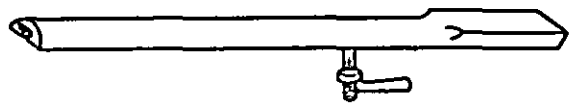


FIGURE 11.—Type XI, hollow mandrel.

3.5.14 *Type XII, candle mold.*—Stakes shall be of forged steel and shall consist of a body and a shank.

3.5.14.1 *Body.*—The body shall be composed of two horns. The tops of the horns shall be level and in the same plane. Both horns shall be circular in cross section, shall be tapered from the shank to the ends, and shall be heat-treated to indicate a Rockwell hardness of not less than 45 nor more than 55 on the C scale.

3.5.14.2 *Shank.*—The shank shall extend from the juncture of the two horns.

3.5.14.3 The stakes shall conform to table XII and shall be similar to figure 12.

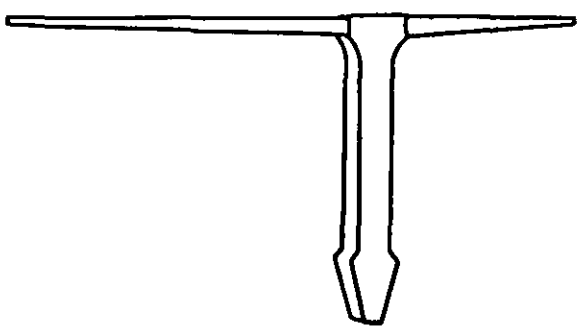


FIGURE 12.—Type XII, candle mold.

TABLE XII.—*Type XII, candle mold*

Length of long horn ±¾	Length of short horn ±⅛	Diameter of long horn at smallest part ±⅛	Diameter of long horn at largest part ±⅛
Inches	*Inches*	*Inch*	*Inch*
18¼	9⅛	⅛	⅛

Diameter of short horn at smallest part ±1/16	Diameter of short horn at largest part ±5/16	Weight ±¾
Inch	*Inches*	*Pounds*
7/16	1 3/16	7¾

3.6 **Type XIII, bench plate.**—Bench plate shall be of close grained cast iron and shall be provided with smooth cored holes tapered to fit standard stake shanks. Smaller holes shall be furnished in the plate for bench shears. The top of the plate shall be machined smooth. Holes shall be provided for bolting the plate to a bench. Bench plate shall conform to table XIII and shall be similar to figure 13.

TABLE XIII.—*Type XIII, bench plate*

Length ±⅛	Width ±¼
Inches	*Inches*
30⅛	8¼



FIGURE 13.—Type XIII, bench plate.

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3.7 Workmanship.—Workmanship shall be of the highest grade and equal in every respect to the best commercial practice. Stakes and plate shall be free from laps, burrs, cracks, and other defects which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All sheet metal stakes or plates of the same type, and size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for visual and dimensional inspection.*—From each lot of stakes or plates formed in accordance with 4.1.1, sample items shall be taken at random in accordance with Military Standard MIL–STD–105 at inspection level III for lots up to 110 items, and inspection level II for lots containing more than 110 items. Each of the sample items shall be inspected in accordance with 4.2. The acceptable quality level (AQL) shall be 1.5 percent.

4.1.3 *Sampling for lot acceptance test.*—Samples for the tests specified in 4.4.1 shall be taken at random from each lot in accordance with inspection level L–8 of the appendix to Military Standard MIL–STD–105. The acceptable quality level (AQL) shall be 4.0 percent.

4.2 Visual and dimensional inspection.—Each of the sample stakes or plates taken in accordance with 4.1.2 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any stake or plate in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective stakes or plates in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.

4.3.1 *Testing of sample stakes for lot acceptance.*—Each of the stakes taken in accordance with 4.1.3 shall be tested in accordance with 4.4.1 to verify compliance with this specification. Any sample stake which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming stakes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Hardness test.*—Hardness test shall be conducted in accordance with Federal Specification QQ–M–151 and the value thus obtained shall comply with the requirements of 3.5.3, 3.5.5, 3.5.6, 3.5.7, 3.5.8, 3.5.10, 3.5.11 and 3.5.14. Impressions shall be taken on all working surfaces.

4.5 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming items. The Government inspector shall again take and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary requirements.

5.1.1 *Preservation.*—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.1.2 *Packaging.*—Commercial packages will be acceptable.

5.1.3 *Packing.* — The equipment shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest rate of the carrier.

5.1.4 *Marking.*

5.1.4.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name of the contractor, or trade-mark of such known character that the source of supply may be readily determined.

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5.1.4.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

5.2 Military requirements.

5.2.1 *Preparation for shipment.*—Packaging and packing for each level of protection shall be in accordance with Military Specification MIL–P–15424. When specified (see 6.1), fiberboard boxes shall not be used for exterior shipping containers for domestic shipments.

6. NOTES

6.1 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of stake and plate required (see 1.2).

(c) Size of type VI or type XI required (see table VI or table XI).

(d) Level of packaging and packing required (see 5.2.1).

(e) When fiberboard boxes should not be used (see 5.2.1).

6.2 Transportation description.—The applicable transportation description is:

Types I to XII, inclusive:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Truckload minimum weight 30,000 pounds.

Type XIII:

Plates, bench, lapping, layout surface or surface, steel.
Carload minimum weight 30,000 pounds.
Truckload minimum weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—O E T
Navy—Sh A Or Y
Air Force.

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